S2S: Physics-Certified Sensor Data

Four Proven Training Benefits Across Two Independent Datasets

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Abstract

S2S (Sensor-to-Standard) is a physics-based certification framework for human motion sensor data. It verifies that sensor recordings obey physical laws governing human movement — not whether data looks human, but whether it obeys the physics of human motion.

We demonstrate four progressive contributions validated across two independent datasets (WISDM 2019 at 20Hz and PAMAP2 at 100Hz with 3 IMUs):

- Level 1 — Quality Floor: Physics floor recovers 108% of corruption damage on WISDM and confirms cross-dataset on PAMAP2.
- Level 2 — Climbing Mechanism: Curriculum training on physics-tiered data with 46% less data outperforms the clean baseline by +1.03% macro F1 on WISDM.
- Level 3 — Kalman Reconstruction: At 20Hz, Kalman smoothing of BRONZE records adds +0.65% F1 over floor filtering alone. Frequency-dependent finding.
- Level 4 — Kinematic Chain Consistency (headline result): Physics-verified multi-sensor fusion achieves +4.23% F1 over single-sensor baseline on 12-class PAMAP2. Chain consistency filter adds +0.91% beyond naive multi-sensor concatenation.

Level 4 is the strongest result: kinematic chain consistency cannot be faked by independent synthetic data generators. Sensors generated independently fail the cross-sensor timing and correlation checks. This makes S2S the first physics-based anti-synthetic-data certification framework for multi-sensor IMU data.

1. Introduction

Large language models learn from text because words have agreed meanings. Human motion sensor data has no equivalent guarantee. A recording labelled ‘walking’ may contain a dead axis, ADC saturation, or data generated synthetically without physical grounding.

S2S applies the same principle to sensor data that language applies to words: certified data means what it claims to mean. Physics laws are the dictionary.

We make four contributions: a quality floor (Level 1), a climbing mechanism (Level 2), frequency-aware reconstruction (Level 3), and kinematic chain consistency for multi-sensor fusion (Level 4). The headline finding is Level 4: physics consistency across multiple sensors provides a training benefit that naive sensor concatenation cannot achieve, and that synthetic data cannot pass.

Adaptive Calibration Principle

The same physics laws produce different score distributions on different sensors and sample rates. The solution: set the floor at p25 of the clean score distribution per dataset. Physics laws stay fixed. Thresholds adapt. GOLD always means all laws passed, everywhere.

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2. Physical Laws Encoded

Single-Sensor Laws (Levels 1-3)

#	Law	Equation	What it catches
1	Newton's Second Law	$F = ma$ (75ms EMG delay)	Synthetic data missing lagged EMG-accel correlation
2	Segment Resonance	$\omega = \sqrt{(K/I)}$	Tremor at impossible frequency for body segment
3	Rigid Body Kinematics	$a = \alpha \times r + \omega^2 \times r$	Gyro and accel generated independently
4	Ballistocardiography	$F = \rho Q v$	IMU missing cardiac recoil present in PPG
5	Joule Heating	$Q = 0.75 \times P \times t$	Sustained EMG without thermal elevation
6	Motor Control Jerk	$\partial^3 x/\partial t^3 \leq 5000$ m/s³ (≥50Hz)	Robotic or keyframe animation artefacts
7	IMU Consistency	Var(accel) ~ f(Var(gyro))	Accel and gyro from independent generators

Multi-Sensor Chain Laws (Level 4)

#	Law	Measure	What it catches
8	Locomotion Coherence	Dominant freq spread < 2.5Hz across IMUs	Sensors recording different activities
9	Segment Coupling	Pearson r(chest,ankle) > 0.3	Independent synthetic channels
10	Gyro-Accel Coupling	r(accel_mag, gyro_mag) > 0.2 per IMU	Rotation without corresponding acceleration

11	Cross-Sensor Jerk Timing	Ankle leads chest by 0-200ms	Reversed or zero lag = not real heel-strike

3. The Tier System

Tier	Score	Laws	Meaning
GOLD	≥87	All	Pristine. All physics laws passed.
SILVER	75–86	Most	Trusted. Minor deviations within noise.
BRONZE	60–74	Marginal	Passes floor. Candidate for Kalman at ≤20Hz.
RECONSTRUCTED	—	Restored	Kalman-smoothed, re-scored ≥75. Weight 0.5.
REJECTED	<floor	Fails	Removed from training pipeline.

4. Level 1: Quality Floor

WISDM 2019 (20Hz, 62,728 records)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	86.4%	0.6304	50,451
B — Corrupted	35% corruption	85.2%	0.6016	50,451
C — S2S Floor	Physics floor	85.8%	0.6327	27,304

Metric	Value	Significance
Corruption damage (B−A)	−2.88% F1	35% corruption breaks training
S2S recovery (C−B)	+3.11% F1	108% of damage recovered
Net vs clean (C−A)	+0.23% F1	Beats clean with 46% less data

WISDM verdict: ✓ PROVEN. 108% recovery.

PAMAP2 Cross-Validation (100Hz, 13,094 windows)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	91.7%	0.9133	10,985

B — Corrupted	35% corruption	89.7%	0.8786	10,985
C — S2S Floor	Adaptive floor p25=69	89.9%	0.8837	6,225

PAMAP2 verdict: ✓ CONFIRMED. Level 1 replicates on independent dataset at 100Hz.

5. Level 2: Climbing Mechanism (WISDM)

Condition	Description	Acc	F1	n
A — Clean	Clean baseline	86.4%	0.6304	50,451
B — Corrupted	Corrupted 35%	85.2%	0.6016	50,451
C — Floor	S2S filtered	85.8%	0.6327	27,304
D — Weighted	Tier-weighted loss	86.3%	0.6384	27,304
E — Curriculum	GOLD→SILVER→BRONZE	86.7%	0.6407	27,304

Level 2 verdict: ✓ PROVEN. +1.03% F1 vs clean with 46% less data.

Note: Level 2 curriculum requires GOLD-tier anchor records. PAMAP2 at 100Hz produces no GOLD records under current calibration, confirming that adaptive tier boundaries are required for cross-Hz generalisation — documented as future work.

6. Level 3: Kalman Reconstruction

Condition	Description	Acc	F1	n
A — Clean	SILVER reference	85.1%	0.6024	17,689
C — Floor	Floor filtered	84.4%	0.6014	12,383
D — Kalman	Floor + Kalman w=0.5	85.5%	0.6079	13,151

WISDM 20Hz verdict: ✓ PROVEN. +0.65% F1. At 100Hz signal density is sufficient — floor alone recovers damage.

7. Level 4: Kinematic Chain Consistency (Headline Result)

Level 4 introduces four new physics laws that operate across multiple sensors simultaneously. The kinematic chain score measures whether hand, chest, and ankle IMUs tell a consistent biomechanical story. Real human motion produces correlated signals across the kinematic chain. Synthetically generated sensor channels, produced independently, fail these cross-sensor tests.

12-Class Activity Recognition (PAMAP2, 100Hz, 3 IMUs)

Condition	Description	Acc	F1	n
A — Single	Chest IMU only	80.3%	0.7969	11,005
B — Multi Naive	3 IMUs concat, no check	83.3%	0.8308	11,005
C — Chain Filter	3 IMUs + chain filter	84.3%	0.8399	8,503
D — Chain+Weights	Chain + tier weights	84.3%	0.8392	8,503

Metric	Value	Significance
Multi-sensor gain (B−A)	+3.39% F1	3 IMUs add genuine information
Chain filter gain (C−B)	+0.91% F1	Consistency check adds beyond concatenation
Net vs single (D−A)	+4.23% F1	Headline: 4.23% from physics-verified fusion
Windows removed by filter	22.7%	Inconsistent multi-sensor data removed
Chain score avg	62.1	539 GOLD, 4,227 SILVER, 4,732 BRONZE

Level 4 verdict: ✓ PROVEN. Kinematic chain consistency: +0.91% over naive fusion, +4.23% net vs single sensor. This is the strongest result in S2S.

Why This Matters for Synthetic Data Detection

Current synthetic motion data generators (MotionDiffuse, MDM, HumanML3D) produce each sensor channel from the same latent representation but do not simulate physical coupling between body segments. A synthetic walking sequence will have correct per-sensor statistics but incorrect cross-sensor timing. Law 11 (ankle leads chest by 0-200ms in real heel-strike) is particularly sensitive: the 50-100ms biomechanical lag is impossible to fake without full rigid-body simulation.

S2S Level 4 is the first physics-based cross-sensor consistency check for IMU data. It catches what single-sensor checks miss.

8. Full Results Summary

Level	WISDM 20Hz	PAMAP2 100Hz	Status
1 — Floor	108% recovery	+0.51% confirmed	✓ Both datasets
2 — Curriculum	+1.03% F1	No GOLD tier at 100Hz	✓ WISDM proven
3 — Kalman	+0.65% F1	Floor sufficient at 100Hz	✓ Frequency-appropriate
4 — Chain	N/A (single sensor)	+4.23% net, +0.91% filter	✓ PAMAP2 proven
Adaptive floor	floor=60	floor=69	✓ Both datasets

9. Limitations and Discussion

Honest Limitations

- Label noise (semantic corruption) is outside physics scope. Physics cannot verify that a recording labelled ‘walking’ is not actually running.
- Jerk bound (Law 6) requires ≥50Hz. Correctly disabled at 20Hz.
- Level 2 curriculum requires GOLD-tier anchor. No GOLD exists at 100Hz under current calibration — adaptive tier boundaries are future work.
- Level 4 validated on PAMAP2 only. Cross-dataset chain consistency (WISDM has one IMU) requires multi-sensor datasets.
- PAMAP2 subject101 excluded (corrupt download). 8 of 9 subjects used.

Discussion: Adaptive Calibration

Physics laws are universal constants. WISDM at 20Hz and PAMAP2 at 100Hz produce different score distributions not because one dataset is higher quality but because the scorer is calibrated to different signal characteristics. Setting the floor at p25 of the clean score distribution allows the same framework to operate across sensors, sample rates, and domains without retuning the physics. GOLD always means all laws passed, everywhere.

Discussion: Kinematic Chain as Anti-Synthetic Signal

The 50-100ms ankle-to-chest jerk lag during heel-strike is a rigid-body consequence of the human skeleton. It is present in every real walking recording regardless of sensor hardware. It is absent in synthetic data unless the generator explicitly simulates skeletal propagation. This makes Law 11 a reliable, hardware-independent marker of real vs synthetic motion data.

Discussion: Frequency-Dependent Reconstruction

Kalman reconstruction is beneficial at 20Hz where signal noise dominates. At 100Hz the temporal resolution is sufficient that the physics floor alone recovers corruption damage and Kalman smoothing removes genuine motion features. This is a scientific finding: reconstruction tools should be matched to signal frequency, not applied universally.

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10. Conclusion

Level	Result
1 — Floor	WISDM 108% recovery. PAMAP2 confirmed. ✓
2 — Curriculum	+1.03% F1 with 46% less data (WISDM). ✓
3 — Kalman	+0.65% at 20Hz. Floor sufficient at 100Hz. ✓
4 — Chain (headline)	+4.23% net, +0.91% filter (PAMAP2 12-class). ✓
Adaptive calibration	p25 floor. Works across Hz and sensors. ✓
Anti-synthetic	Cross-sensor timing check catches independent generators. ✓

Physics laws are universal constants. Certification thresholds are dataset-relative variables. The kinematic chain cannot be faked without full biomechanical simulation. That is the S2S contribution.

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